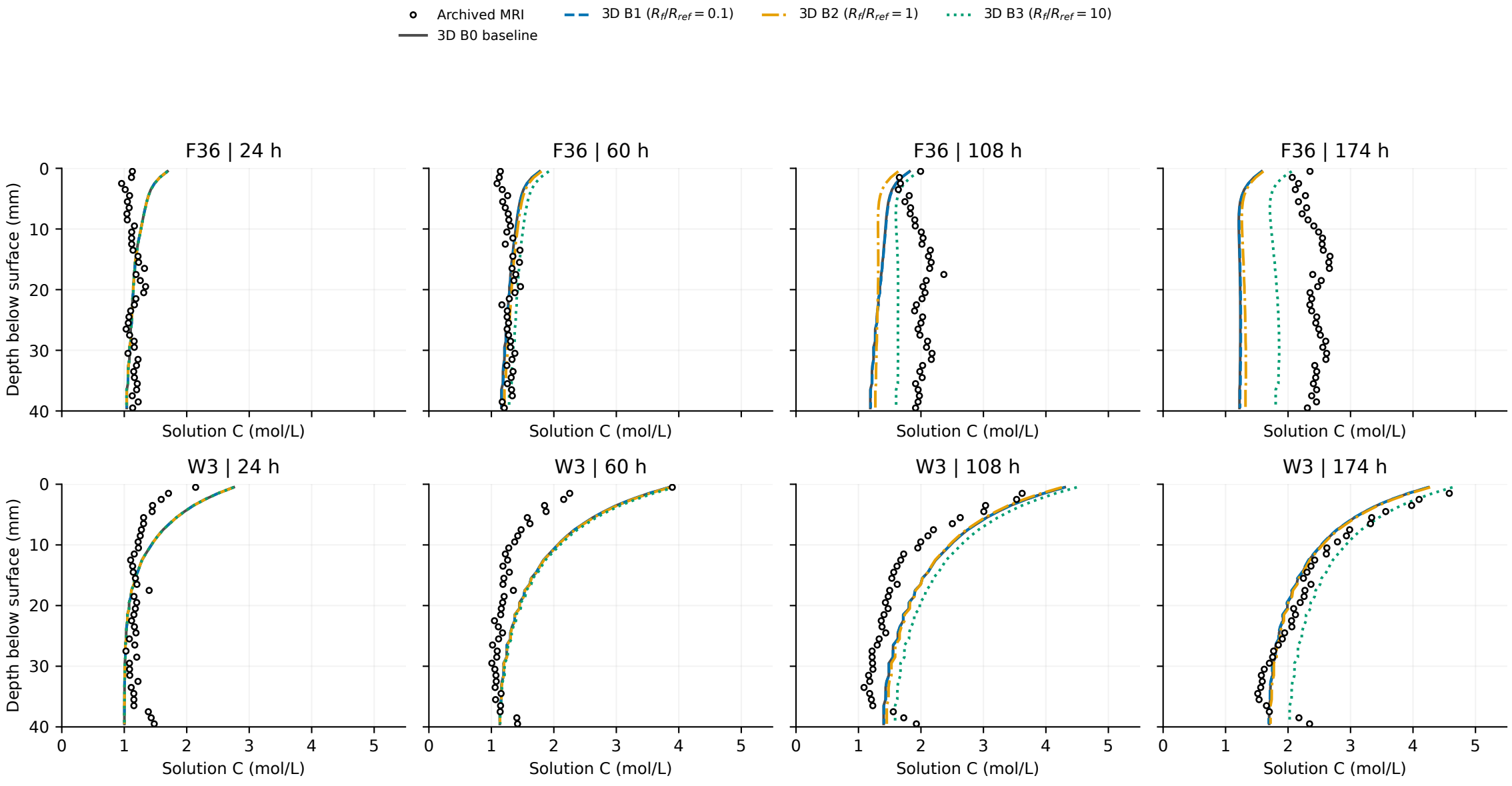
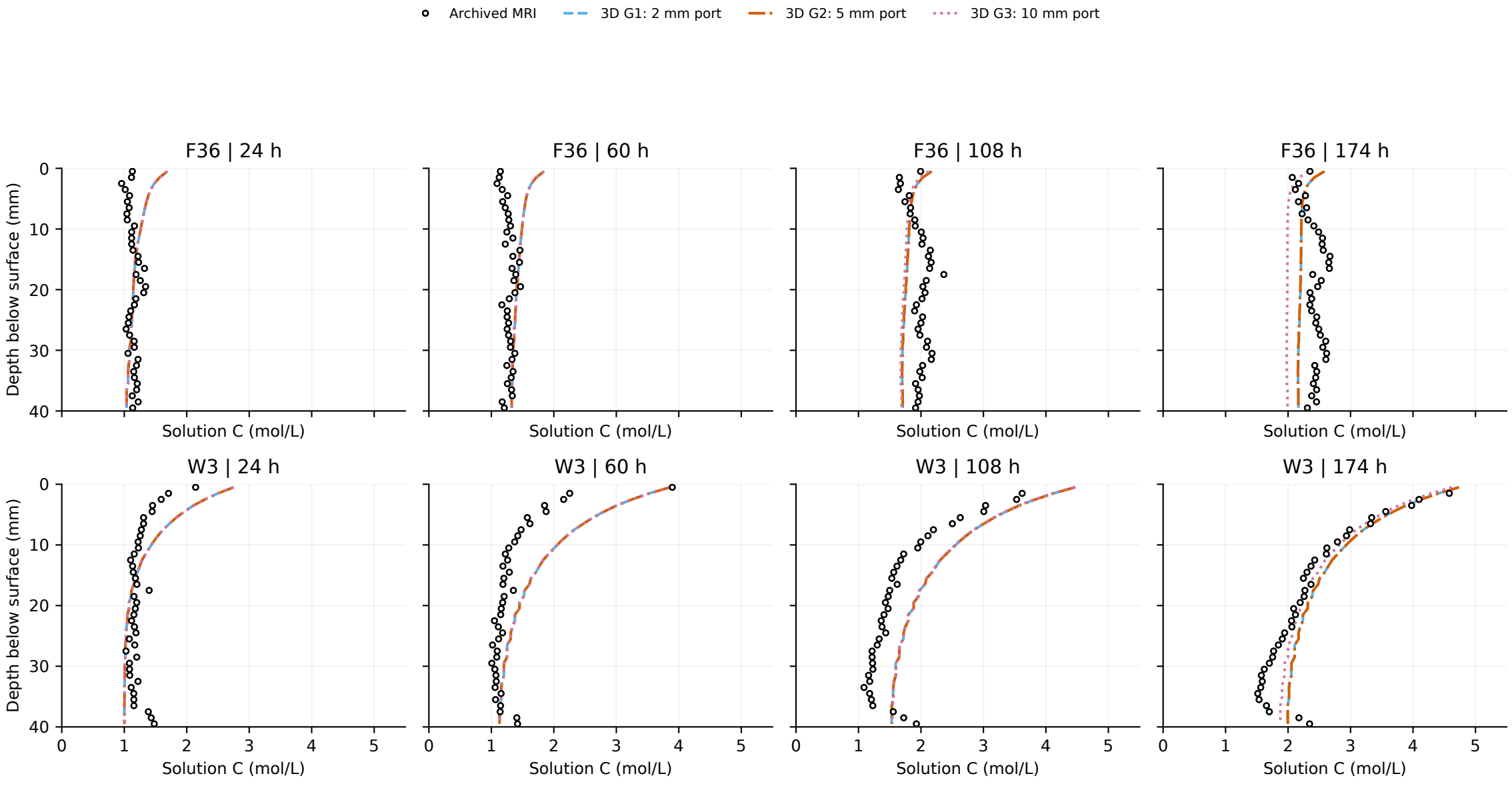
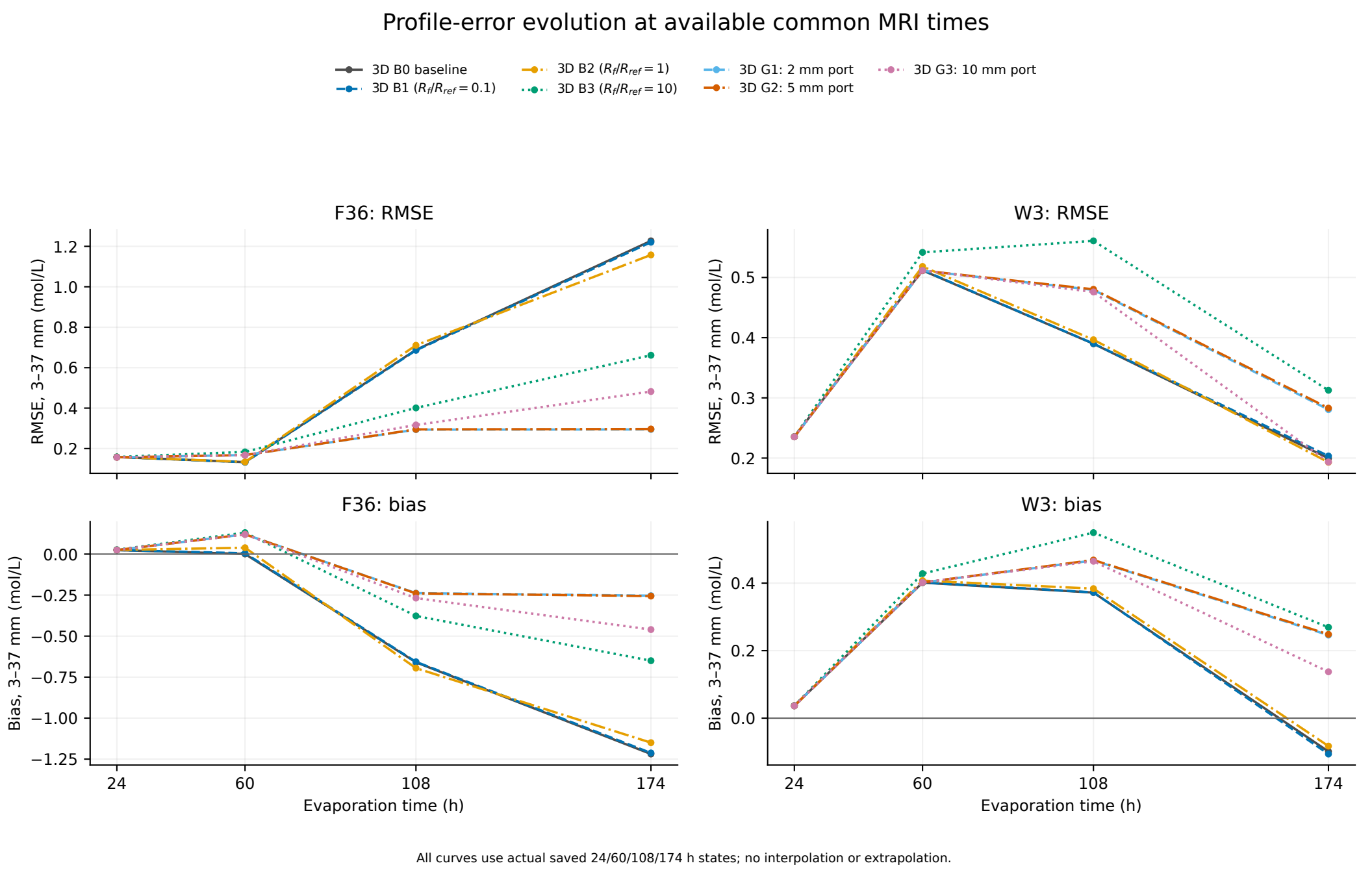
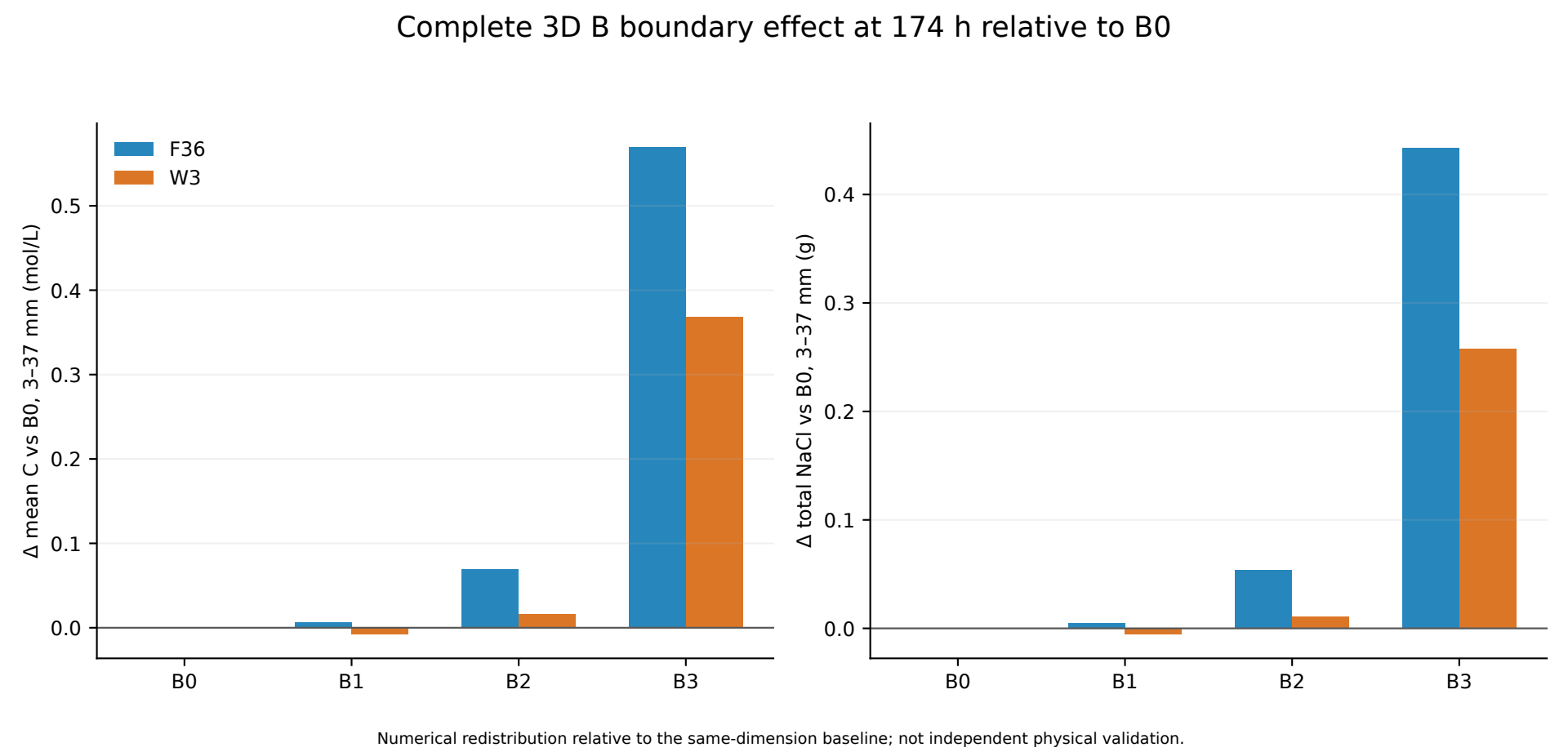
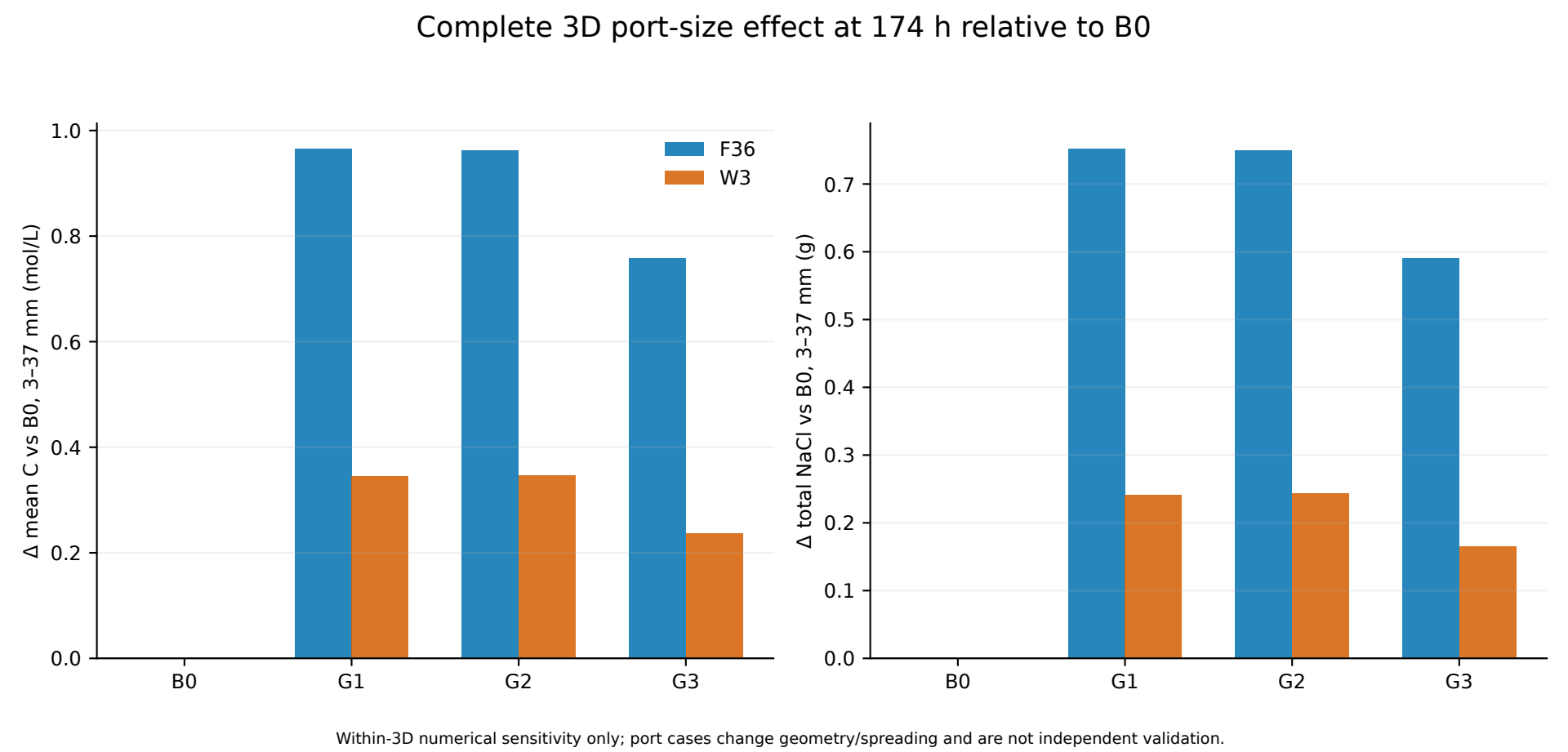


Whole-half profiles; actual saved states and nominal MRI clock; no interpolation or onset inference.


3D G1–G3 complete 174 h results versus archived MRI
Archived MRI
3D G1: 2 mm port
3D G2: 5 mm port
3D G3: 10 mm port
F36 | 24 h
F36 | 60 h
F36 | 108 h
F36 | 174 h
W3 | 24 h
W3 | 60 h
W3 | 108 h
W3 | 174 h
Depth below surface (mm)
Solution C (mol/L)
0
10
20
30
40
0
1
2
3
4
5
Whole-half profiles; actual saved states and nominal MRI clock; no interpolation or onset inference.


Profile-error evolution at available common MRI times
3D B0 baseline
3D B1 ($R_f/R_{ref} = 0.1$)
3D B2 ($R_f/R_{ref} = 1$)
3D B3 ($R_f/R_{ref} = 10$)
3D G1: 2 mm port
3D G2: 5 mm port
3D G3: 10 mm port
F36: RMSE
W3: RMSE
F36: bias
W3: bias
RMSE, 3–37 mm (mol/L)
Bias, 3–37 mm (mol/L)
Evaporation time (h)
24
60
108
174
All curves use actual saved 24/60/108/174 h states; no interpolation or extrapolation.


Complete 3D B boundary effect at 174 h relative to B0
F36
W3
Δ mean C vs B0, 3–37 mm (mol/L)
0.0
0.1
0.2
0.3
0.4
0.5
B0
B1
B2
B3
Δ total NaCl vs B0, 3–37 mm (g)
0.0
0.1
0.2
0.3
0.4
B0
B1
B2
B3
Numerical redistribution relative to the same-dimension baseline; not independent physical validation.


Complete 3D port-size effect at 174 h relative to B0
Δ mean C vs B0, 3–37 mm (mol/L)
Δ total NaCl vs B0, 3–37 mm (g)
F36
W3
B0
G1
G2
G3
Within-3D numerical sensitivity only; port cases change geometry/spreading and are not independent validation.